

Beyond Single-Pair Attacks: Disrupting Vision-Language Pre-Training Models With Dual-Semantic Frequency Stealth

Haiqi Zhang¹, Ziqiang Li², Hao Tang³, *Member, IEEE*, and Zechao Li¹, *Senior Member, IEEE*

Abstract—Vision-Language Pre-training (VLP) models are highly capable in multimodal tasks but are critically vulnerable to adversarial attacks. Existing methods for creating transferable adversarial examples typically operate by modifying semantics within isolated image-text pairs. This strategy often yields poorly generalizable perturbations that insufficiently disrupt cross-modal alignment, leading to limited effectiveness in black-box settings. An additional limitation is the frequent oversight of adversarial stealth. Toward this end, we propose the **Dual-Semantic Frequency-Guided Transferable Attack (DSFG-Attack)**, which simultaneously disrupts semantic consistency through cross-modal interaction diversification and enforces perturbation stealth via frequency-domain constraints. Specifically, our method (1) injects non-aligned semantics into image-text pairs to break inherent multimodal correlations and (2) restricts perturbations to high-frequency domains (e.g., fine textures) for enhanced concealment. Extensive experiments on image-text retrieval show that DSFG-Attack significantly improves transferability, elevating the attack success rate by an average of 5.08% in black-box settings (source model: TCL; target model: CLIP_{ViT}; dataset: Flickr30K) and also strengthening white-box performance. The adversarial examples further enable cross-domain adversarial transfers, including both cross-task (retrieval → captioning) and cross-model (VLP → Multimodal Large Language Models), thereby inducing semantic misalignment in advanced systems like GPT-4o and Qwen 2. Notably, evaluations across multiple metrics confirm a significant enhancement in the stealthiness of the crafted adversarial examples.

Index Terms—Image-text retrieval, multi-modal black-box attack, vision-language pre-training models, multi-modal large language models.

I. INTRODUCTION

VISION-LANGUAGE Pre-training (VLP) models have become foundational components in multimodal artificial intelligence, demonstrating exceptional performance on tasks

such as cross-modal retrieval, visual question answering, and image captioning [1], [2], [3]. Their ability to fuse and reason over visual and linguistic information has enabled significant advancements. However, this capability comes with a critical and often overlooked vulnerability: like their unimodal counterparts, VLP models are highly susceptible to adversarial perturbations [4], [5], [6]. Minute, carefully crafted alterations to input images or text can lead to catastrophic failures in model predictions. This fragility poses a substantial security threat, casting doubt on the reliability of deploying these systems in safety-critical real-world applications and necessitating a rigorous investigation into their adversarial robustness [7], [8], [9].

Research on attacking VLP models has evolved along two primary axes defined by the attacker's knowledge [10]. White-box attacks assume full access to the model's architecture and parameters, allowing for precise, gradient-based optimization of adversarial perturbations. In contrast, more practical black-box attacks operate under a constrained setting, where the attacker can only query the target model and must rely on the transferability of adversarial examples crafted on a surrogate model [11], [12]. Early work (such as Co-Attack) demonstrated the feasibility of white-box attacks by jointly optimizing perturbations across fused multimodal features [13]. However, its dependence on internal model access limits real-world applicability. Consequently, recent efforts have shifted towards improving black-box transferability. Methods like SGA perturb the cross-modal alignment within a surrogate model using augmented data distributions, and its variant DRA refines this process through strategic interpolation to construct perturbation regions [14], [15].

Despite these advancements, a fundamental limitation persists. Current methods primarily operate by maximizing the semantic divergence between a single adversarial image-text pair, as learned by the source (surrogate) model [16], [17]. They inadequately address a core challenge of transferability: how to ensure that this induced semantic misalignment generalizes and remains potent when transferred to a different, unknown target model with potentially disparate feature representations and decision boundaries. This often results in a significant performance gap between white-box and black-box attack success rates. Furthermore, existing work largely neglects the critical aspect of adversarial example stealthiness, a paramount requirement for practical, real-world attacks where detectable anomalies would raise immediate suspicion.

Received 2 November 2025; revised 31 May 2026; accepted 11 July 2026. Date of publication 14 July 2026; date of current version 1 September 2026. This work was supported in part by the National Natural Science Foundation of China under Grant 62425603 and in part by the Basic Research Program of Jiangsu Province under Grant BK20240011. (Corresponding author: Zechao Li.)

Haiqi Zhang and Zechao Li are with the School of Computer Science and Engineering, Nanjing University of Science and Technology, Nanjing 210094, China (e-mail: zhanghq2017@njust.edu.cn; zechao.li@njust.edu.cn).

Ziqiang Li is with the School of Computer Science, Nanjing University of Information Science and Technology, Nanjing 210044, China (e-mail: iceli@mail.ustc.edu.cn).

Hao Tang is with the Centre for Smart Health, The Hong Kong, Polytechnic University, Hong Kong, SAR, China (e-mail: howard.haotang@gmail.com).

Digital Object Identifier 10.1109/TDSC.2026.3713210

To simultaneously address these dual challenges of generalizable transferability and practical stealth, we propose the **Dual-Semantic Frequency-Guided Transferable Attack (DSFG-Attack)**. Our core insight is that merely pushing an adversarial example away from its original counterpart in the source model's embedding space creates a fragile, model-specific divergence. To forge a more robust and transferable perturbation, DSFG-Attack introduces a novel dual-semantic guidance mechanism. This mechanism actively steers the adversarial generation by enforcing two concurrent objectives: (1) repelling the adversarial example from the semantic region of its originally paired modality, and (2) attracting it toward the semantic neighborhood of a carefully chosen, dissimilar data point. This strategy ensures the perturbation carries a coherent, yet deliberately misdirected, semantic signal that is more likely to be recognized as a discrepancy across diverse VLP model architectures, thereby significantly enhancing cross-model transferability.

Concurrently, DSFG-Attack incorporates explicit constraints to ensure the stealth of the generated examples. For the image modality, we formulate the attack in the frequency domain, concentrating perturbations within high-frequency components that are less perceptible to the human visual system, thereby preserving visual fidelity. For the text modality, we perform perturbations directly within the word embedding space, enabling semantically coherent substitutions that maintain grammatical integrity and avoid the nonsensical outputs typical of aggressive token-replacement strategies.

Without loss of generality, we choose the standard multi-modal datasets Flickr30K [18] and MSCOCO [19] to evaluate DSFG-Attack on the image-text retrieval task. Extensive experimental analysis demonstrates that DSFG-Attack significantly enhances the semantic differences between modalities and effectively mitigates the performance gap in attacks transitioning from white-box to black-box scenarios. In particular, DSFG-Attack improves task performance in the black-box scenario and also exhibits a strong attack effect in the white-box attack task setting. For example, in the Text-to-Image Retrieval task, when CLIP_{ViT} [20] is used as the source model to generate adversarial examples from the Flickr dataset, the success rate of attacks on the model itself (white-box attacks) increases by an average of 1.04%. Additionally, the success rate of attacks on the CLIP_{CNN} model (black-box attacks) improves by an average of 6.50% compared to the latest methods. Quantitative analysis using multiple perceptual metrics (including ℓ_2 norm [21], SSIM [22], CIEDE2000 [23], and LF [24]) confirms the visual stealth of our adversarial images. We further validate the method's efficacy in more challenging cross-task (retrieval $\rightarrow$ captioning [25]) and cross-model (VLP model $\rightarrow$ Multimodal Large Language Model [26], [27]) scenarios, providing broader insights into multimodal robustness.

Our contributions are summarized as follows:

- **Dual-Semantic Guided Attack Framework:** We propose a dual-semantic guidance mechanism to facilitate transferable multimodal adversarial attacks. By actively attracting examples towards a non-similar semantic anchor, our method effectively disrupts cross-modal semantic alignment, thereby narrowing the performance gap between

white-box and black-box attack scenarios. **This strategy also contributes to improved stealthiness of the adversarial images.**

- **Frequency-Constrained Image Perturbation:** We introduce a frequency-domain constraint for image attacks, optimized jointly with the dual-semantic objective. This confines perturbations to less perceptible high-frequency regions, ensuring strong visual stealth while maintaining tight control over perturbation magnitude.
- **Semantic-Preserving Text Perturbation:** We develop a text perturbation strategy that operates in the embedding space. This approach restricts word-level modifications to nearby semantic regions, minimizing lexical distortion and preserving grammatical and semantic coherence, thereby avoiding nonsensical outputs commonly produced by aggressive replacement strategies.
- **Extensive Empirical Validation:** Through comprehensive experiments with multiple quantitative metrics, we demonstrate that DSFG-Attack significantly enhances the **stealthiness** of adversarial examples. It maintains improved attack performance across diverse VLP models and also in cross-task and cross-model scenarios. These results confirm its capability to achieve robust performance with minimal, highly stealthy perturbations.

II. RELATED WORK

A. Vision-Language Pre-Training Models

Vision-language pre-training (VLP) models leverage extensive image-text pairs to acquire inter-modal associative semantic information, substantially enhancing their performance on multi-modal tasks [28], [29]. Initially, the field heavily relied on pre-trained object detectors for vision-language representations, a method requiring significant computational resources [30], [31], [32]. However, with the advent and adoption of vision transformers (ViT), vision-language representation learning methods have undergone a transformation. Researchers advocate replacing the pre-trained object detector with ViT to expedite the end-to-end model inference process [33], [34], [35].

According to the model structure, VLP models can be categorized into two types: fused VLP models and aligned VLP models [36], [37]. The fused VLP model, exemplified by AL-BEF [38] and TCL [39], comprises three encoders: two unimodal encoders and one multi-modal encoder. In this model, the unimodal image and text encoders initially encode the image-text pairs to produce their respective image and text embedding representations. Subsequently, the multi-modal encoder integrates these embeddings to generate a unified representation for the image-text pairs. In contrast, the aligned VLP model consists of only two unimodal encoders. It directly utilizes the embedding representations of these two unimodal encoders to perform downstream tasks without undergoing a multi-modal fusion stage [20], [40].

B. Image-Text Retrieval Task

Image-text retrieval is one of the common downstream tasks of the VLP model, which is further divided into two subtasks:

image-to-text retrieval and text-to-image retrieval [41], [42]. The implementation of the image-text retrieval task varies significantly between the two types of VLP models. In the fused VLP model, two unimodal encoders are initially employed to produce separate embedded representations for all image-text pairs. Then, the top- n candidates are selected based on the inter-modal feature similarity scores, and the image-text matching scores are computed using the multi-modal encoders for ranking. On the contrary, in the aligned VLP model without multi-modal encoders, the image-text embedded representations obtained from the unimodal encoders are directly used to compute the similarity to arrive at the final retrieval results.

C. Adversarial Attack

1) *Unimodal Adversarial Attack Methods*: Unimodal adversarial attacks can be broadly categorized into two types: image attacks and text attacks. Image attacks typically employ gradients to generate perturbations by optimizing the loss function of target model's output decision boundary [43], [44], [45], [46], [47]. Meanwhile, to ensure that the perturbations are imperceptible to the human eye and do not significantly alter the original image, they are constrained within a small interval. In contrast, text attacks are more intricate [48], [49]. Text data occupies a discrete space, rendering attacks based on gradients impractical and presenting a challenging obstacle. Currently, the most prevalent method for text attacks is the BERT attack, which enhances error output by modifying the text through word removal, addition, or replacement [50], [51].

2) *Multi-Modal Adversarial Attack Methods*: Multimodal adversarial attacks remain less explored compared to their unimodal counterparts [52], [53], [54], [55]. Recent methods such as Sep-Attack, Co-Attack [13], SGA [14], VLATTACK [36], SA-Attack [56], and DRA [15] have investigated the robustness of VLP models. Among these, Sep-Attack shows notable limitations: it relies solely on single sample pairs for perturbation learning, leading to overfitting to the source model and poor black-box transferability. Moreover, it applies unimodal techniques (PGD [43] for images, BERT-Attack [50] for text) without cross-modal interaction, failing to disrupt the image-text alignment essential to VLP downstream tasks, thereby limiting attack transferability.

Co-Attack introduces one round of interaction per pair: it first perturbs text using BERT-Attack, then uses the adversarial text to guide image perturbation generation. However, it still depends on single-pair information, which inadequately disrupts cross-modal alignment, and its text perturbations receive no guidance from image content, resulting in weak semantic interaction.

SGA enhances interaction via set-level augmentation, increasing the number of cross-modal interactions to two. Although it diversifies image data, it remains constrained by single-pair semantics. Although DRA extends SGA by replacing point sampling with triangular region sampling, the approach remains insufficient for preserving semantic divergence in black-box settings.

To address these issues, we propose leveraging semantics from dissimilar samples to guide adversarial perturbation generation, aiming to enhance transferability in multimodal

adversarial attacks. Furthermore, for the first time from the perspective of adversarial stealth in multimodal settings, we impose frequency-domain constraints on images and lexical constraints on textual perturbations to ensure the perceptual quality and transferability of the attacks.

III. PRELIMINARIES

This section establishes the foundational concepts and formal definitions underpinning our work. We begin by introducing the key notations and model paradigms relevant to VLP models. Subsequently, we formalize the objectives and categorize the scenarios of multimodal adversarial attacks. Finally, we present a core theoretical insight on semantic relationship preservation across models, which motivates the design of our proposed attack methodology.

A. Notations and VLP Model Architectures

We standardize the notation used throughout this paper. VLP models primarily follow two architectural paradigms based on their fusion mechanisms [36]:

- *Fusion-based VLP Models*: These consist of three core components: a unimodal image encoder E_I , a unimodal text encoder E_T , and a multimodal fusion encoder E_M . The fusion encoder jointly processes information from both modalities.
- *Alignment-based VLP Models*: These models lack a dedicated fusion encoder E_M . They rely solely on the separate encoders E_I and E_T , and perform cross-modal matching directly in a shared or aligned embedding space.

For an input image i and an input text sequence t , we denote their corresponding unimodal embeddings as $E_I(i)$ and $E_T(t)$, respectively. In fusion-based models, the final multimodal representation is given by $E_M(E_I(i), E_T(t))$.

B. Threat Model: Attack Objectives and Scenarios

We formalize the threat model for multimodal adversarial attacks against VLP models. The overarching goal is to craft imperceptible perturbations (δ_i, δ_t) that, when added to a clean image-text pair (i, t) , cause a significant deviation in the model's output representation, thereby degrading performance on downstream tasks such as retrieval or classification.

Formally, for a fusion-based model, a successful attack aims to satisfy:

$$E_M(E_I(i), E_T(t)) \neq E_M(E_I(i \oplus \delta_i), E_T(t \oplus \delta_t)), \quad (1)$$

where $\oplus$ denotes the adversarial perturbation operation, the specifics of which may differ between the image and text modalities (e.g., additive noise for images, word substitutions for text).

To reflect practical constraints, we consider two primary attack scenarios based on the attacker's knowledge:

- *White-box Attacks*: The attacker has full access to the target model, including its architecture, parameters, and gradients. This setting allows for precise, gradient-based optimization of adversarial perturbations, typically yielding high attack success rates.

- **Black-box Attacks:** The attacker has no access to the internal workings of the target model. Attacks must be generated using a surrogate model and rely on the transferability of adversarial examples across different models. This scenario is more challenging but aligns with real-world deployment conditions.

Our work primarily focuses on the more practical and challenging black-box setting.

C. Theoretical Motivation: Cross-Model Semantic Preservation

A key insight motivating our method is the observation that semantic relationships between data points exhibit a degree of preservation across different VLP models, even if their absolute embeddings differ. Specifically, consider two highly dissimilar images (i_m, i_j) with a cosine similarity $s_{m,j} < \sigma$ (where σ is a small threshold) in the embedding space of a white-box surrogate model with encoder E_I .

Theory from model manifold alignment [57] suggests that the relative semantic distance between such dissimilar pairs tends to be preserved, or at least not inverted, in the embedding space of a black-box target model with encoder E_I^t . This can be expressed as a bounded dissimilarity preservation:

$$\|E_I^t(i_m) - E_I^t(i_j)\|_2 \geq \gamma \|E_I(i_m) - E_I(i_j)\|_2, \quad (2)$$

where $\gamma \in (0, 1)$ is a factor quantifying the strength of this geometric preservation across models.

This property provides a crucial theoretical foundation for our attack strategy. If an adversarial image i_m^{adv} is actively guided towards the semantic region of a dissimilar image i_j in the surrogate model (making $\|E_I(i_m^{adv}) - E_I(i_j)\|_2$ small), the inequality above implies that i_m^{adv} is likely to remain semantically closer to i_j than to its original counterpart in the target model as well. This promotes the transferability of the induced semantic shift, making the adversarial example effective against unseen models. Our proposed DSFG-Attack directly leverages this principle by using a dissimilar image as a semantic anchor to guide the generation of transferable adversarial perturbations.

IV. METHODOLOGY

This section presents the **Dual-Semantic Frequency-Guided Attack (DSFG-Attack)**, a transferable multimodal adversarial attack against VLP models. DSFG-Attack aims to induce robust image-text semantic misalignment under black-box settings, where the target model's parameters and gradients are inaccessible. The framework contains two main designs: 1) *dual-semantic guidance*, which introduces semantics from a non-similar sample to enlarge cross-modal discrepancy beyond a single image-text pair; and 2) a *frequency-domain constraint*, which confines image perturbations to less perceptible high-frequency regions for improved stealth.

A. Overview

As discussed in Section II-C2, adversarial examples generated from isolated image-text pairs often overfit the source model's

alignment pattern and thus transfer poorly to unseen models. To address this issue, DSFG-Attack leverages a non-corresponding and semantically dissimilar sample as an additional guidance signal, encouraging a more transferable semantic shift.

As shown in Fig. 1, DSFG-Attack consists of two sequential stages.

- **Image-Guided Text Attack:** Given an image-caption pair, we identify the semantically critical word in the caption and replace it with a plausible candidate that minimizes its similarity with the paired image. This produces adversarial captions that remain linguistically natural but weaken image-text alignment.
- **Text-Guided Image Attack:** Using the adversarial captions as guidance, we perturb the image such that its representation is repelled from the adversarial text while being attracted toward a non-similar image selected from the same mini-batch. This dual-semantic guidance introduces a coherent yet incorrect semantic drift, improving black-box transferability.

To enhance visual stealth, we further introduce a frequency-domain constraint during image perturbation. By penalizing changes in low-frequency components, DSFG-Attack encourages perturbations to concentrate on high-frequency textures, which are less perceptible to the human visual system [24].

B. Dual-Semantic Frequency-Guided Attack

1) **Image-Guided Textual Adversarial Generation:** The first stage crafts adversarial perturbations in the textual modality using the paired image as semantic guidance. Given a mini-batch of M image-caption pairs $\{(i_1, t_1), (i_2, t_2), \dots, (i_M, t_M)\}$, each image i_m is associated with N captions $t_m = (t_{m1}, t_{m2}, \dots, t_{mN})$. Our goal is to generate an adversarial caption set $t_m^{adv} = (t'_{m1}, t'_{m2}, \dots, t'_{mN})$ whose semantic similarity with i_m is minimized in the joint embedding space, thereby weakening the original image-text correspondence.

For each caption $t_{mn} = [a_1, a_2, \dots, a_L]$, the adversarial generation consists of three steps.

Step 1. Localizing Semantically Critical Tokens: We identify the token most responsible for image-text alignment. Specifically, each token a_l is masked in turn, producing a set of masked captions $\{t_{mn,l}^{mask}\}_{l=1}^L$. Each masked caption is encoded by E_T , and its cross-modal importance score s_l is computed with the fixed image embedding $E_I(i_m)$. The token whose masking yields the highest importance score is selected as the vulnerable lexical unit for replacement.

Step 2. Generating Semantically-Plausible Candidates: To avoid unnatural substitutions, we search for replacement candidates in a semantic neighborhood of the selected word. Let e_{a_l} denote the embedding of the selected token a_l . Using a pre-trained word embedding space such as GloVe [58], [59], we define the candidate set as

$$C_{mn} = \{c \mid \text{sim}(e_c, e_{a_l}) \geq \gamma\}, \quad (3)$$

where $\text{sim}(\cdot, \cdot)$ denotes cosine similarity and γ controls semantic proximity. This constraint helps preserve the local syntactic and semantic plausibility of the caption.

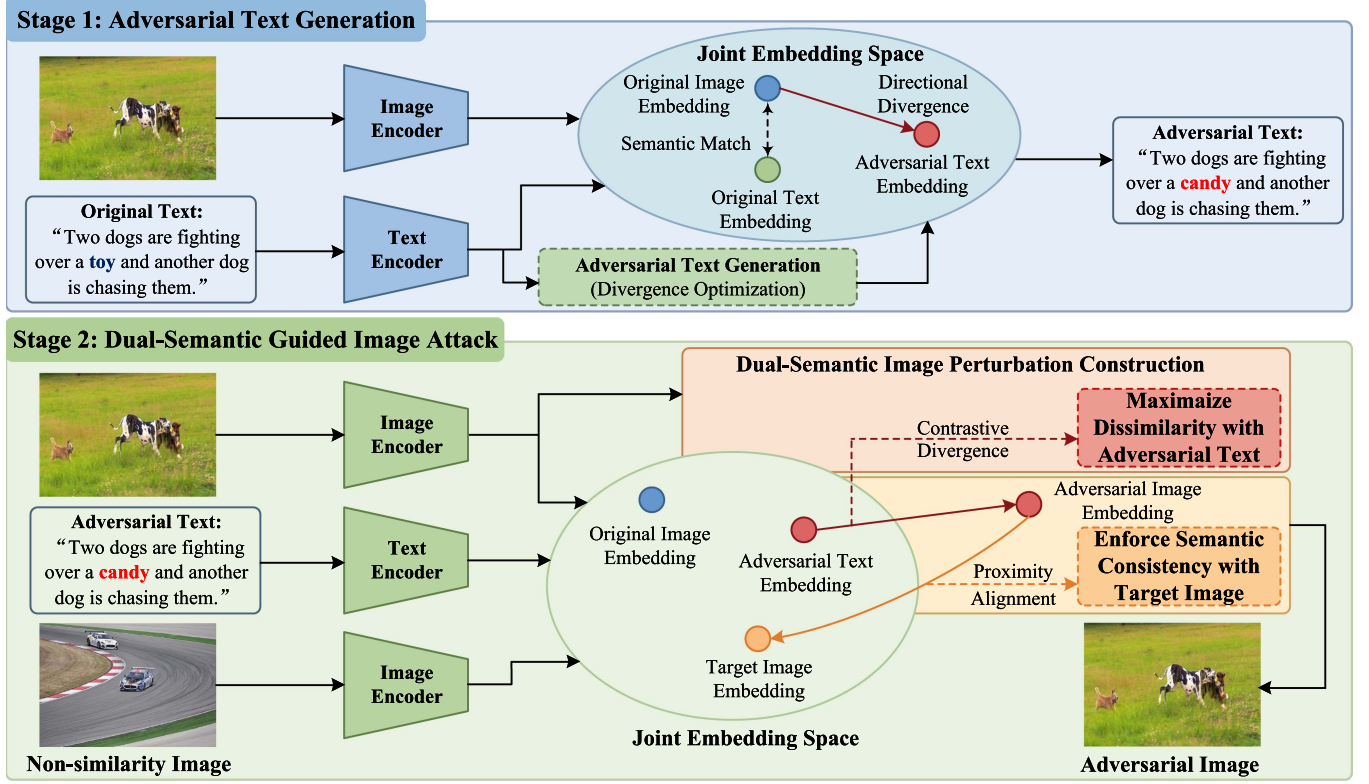


Fig. 1. Description of DSFG-Attack. **Stage 1: Adversarial Text Generation.** Guided by the original image’s visual semantics, we generate text perturbations in the joint embedding space. Through directional divergence optimization, the adversarial text embeddings are systematically displaced from the original image semantics to disrupt cross-modal alignment. **Stage 2: Dual-Semantic Guided Image Attack.** Leveraging both adversarial text embeddings and predefined target (non-similarity) semantics, we construct image perturbations that simultaneously: (1) maximize dissimilarity with the adversarial text through contrastive divergence, and (2) enforce semantic consistency with designated target features via proximity alignment in the latent space.

Step 3: Calibrating for Maximal Cross-Modal Deviation: From C_{mn} , we select the candidate that most reduces the similarity between the image and the substituted caption:

$$t'_{mn} = \arg \min_{c \in C_{mn}, t'_{mn} \in S[t_{mn}, \varepsilon_t]} \frac{E_I(i_m) \cdot E_T(\hat{t}_{mn}(c))}{\|E_I(i_m)\| \|E_T(\hat{t}_{mn}(c))\|}, \quad (4)$$

where $\hat{t}_{mn}(c)$ denotes the caption obtained by replacing a_l with candidate c . The perturbation space $S[t_{mn}, \varepsilon_t]$ limits the number of word substitutions, with $\varepsilon_t = 1$ in our setting to ensure textual stealth. This produces adversarial captions that remain plausible while effectively weakening their semantic alignment with the paired image.

2) *Text-Guided Adversarial Image Generation:* Given the adversarial caption set t_m^{adv} , we generate adversarial images using dual-semantic guidance and a frequency-domain constraint. The former enhances transferable semantic misalignment, while the latter restricts perturbations to less perceptible frequency regions.

a) *Dual-Semantic Guidance:* For each source image i_m , we synthesize an adversarial image i_m^{adv} that is repelled from the adversarial text and attracted toward a semantically dissimilar image. This process is conducted within a mini-batch $\mathcal{I} = \{i_1, i_2, \dots, i_M\}$ and consists of target selection, similarity assessment, and integrated optimization.

Step 1: Non-Similar Target Selection: We select a non-similar image from the mini-batch as the semantic anchor. Specifically, the cosine similarity between i'_m (optimization variable, initialize $i'_m \leftarrow i_m$) and each other image i_j ($j \neq m$) is computed in the image embedding space:

$$s_{m,j} = \frac{E_I(i'_m) \cdot E_I(i_j)}{\|E_I(i'_m)\| \|E_I(i_j)\|}, \quad \forall j \neq m. \quad (5)$$

The image with the minimum similarity is denoted as i_{ns} and used as the non-similar target. This anchor guides the adversarial image away from its original semantic neighborhood toward a distinct visual semantic region.

Step 2: Dynamic Similarity Assessment: During the iterative generation of an intermediate adversarial image i'_m , we evaluate its semantic relationships with two references: the adversarial caption set $t_m^{adv} = \{t'_{m1}, t'_{m2}, \dots, t'_{mN}\}$ and the selected non-similar target i_{ns} . Accordingly, we define two scores.

- *Text-Image Divergence Score (D_T):* This score measures the alignment between i'_m and its adversarial captions:

$$D_T(i'_m) = \frac{E_I(i'_m)}{\|E_I(i'_m)\|} \cdot \sum_{n=1}^N \frac{E_T(t'_{mn})}{\|E_T(t'_{mn})\|}. \quad (6)$$

- *Image-Image Attraction Score (A_I):* This score measures the semantic affinity between i'_m and the non-similar

Algorithm 1: Image-Guided Textual Adversarial Generation.

Input: Mini-batch $\{(i_m, t_m)\}_{m=1}^M$, each with N captions; encoders E_I, E_T ; similarity threshold γ ; perturbation budget ε_t .

- 1: **for** $m = 1$ **TO** M **do**
- 2: **for** $n = 1$ **TO** N **do**
- 3: Let $t_{mn} = [a_1, \dots, a_L]$.
- 4: **for** $l = 1$ **TO** L **do**
- 5: Mask token a_l and compute its cross-modal importance score.
- 6: **end for**
- 7: Select the most critical token a_l .
- 8: Retrieve candidate set
- 9: $C_{mn} = \{c \mid \text{sim}(\mathbf{e}_c, \mathbf{e}_{a_l}) \geq \gamma\}$ from GloVe.
- 10: Select replacement:
- 11: $t'_{mn} = \arg \min_{c \in C_{mn} \cap t'_{mn} \in S[t_{mn}, \varepsilon_t]} \frac{E_I(i_m) \cdot E_T(t'_{mn}(c))}{\|E_I(i_m)\| \|E_T(t'_{mn}(c))\|}$
- 12: **end for**
- 13: Collect $t_m^{adv} = (t'_{m1}, \dots, t'_{mN})$.
- 14: **end for**

Output: Adversarial caption sets $\{t_m^{adv}\}_{m=1}^M$.

target i_{ns} :

$$A_I(i'_m) = \frac{E_I(i'_m) \cdot E_I(i_{ns})}{\|E_I(i'_m)\| \|E_I(i_{ns})\|}. \quad (7)$$

The non-similar image i_{ns} is not used as an instance-level adversarial label. Instead, it acts as a sample-adaptive semantic anchor that provides a coarse direction for semantic displacement. Since the anchor varies across samples and mini-batches, the adversarial image is not optimized to imitate i_{ns} . Rather, it is guided away from the original image-text semantic neighborhood toward a distinct visual region. This large-scale semantic displacement helps reduce dependence on source-model-specific gradients and improves transferability.

Step 3: Integrated Dual-Semantic Optimization: The final adversarial image is obtained by jointly encouraging repulsion from the adversarial text and attraction to the non-similar visual anchor:

$$i_m^{adv} = \arg \min_{i'_m} [\lambda D_T(i'_m) - (1 - \lambda) A_I(i'_m)]_+, \quad (8)$$

where $\lambda \in (0, 1)$ balances the two guidance signals, and $[\cdot]_+$ keeps the loss non-negative.

Design Rationale and Advantage: Minimizing $D_T(i'_m)$ directly weakens the image-text alignment, while maximizing $A_I(i'_m)$ induces a coherent semantic drift toward an unrelated visual region. Compared with attacks that only break the text-image correspondence, this dual guidance re-anchors the adversarial image to a meaningful but incorrect semantic area, making the perturbation less source-model-specific and more transferable across architectures, as further analyzed in Section III-C.

Algorithm 2: Text-Guided Adversarial Image Generation with Frequency Constraint.

Input: Images $\{i_m\}_{m=1}^M$; adversarial captions $\{t_m^{adv}\}_{m=1}^M$; encoder E_I, E_T ; iteration count K ; balancing coefficient λ ; frequency constraint weight β .

- 1: **for** $m = 1$ **TO** M **do**
- 2: Initialize $i'_m \leftarrow i_m$.
- 3: **repeat**
- 4: Select non-similar target:
- 5: $i_{ns} = \arg \min_{i_j, j \neq m} \frac{E_I(i'_m) \cdot E_I(i_j)}{\|E_I(i'_m)\| \|E_I(i_j)\|}$
- 6: Compute text-image divergence:
- 7: $D_T(i'_m) = \frac{E_I(i'_m)}{\|E_I(i'_m)\|} \cdot \sum_{n=1}^N \frac{E_T(t'_{mn})}{\|E_T(t'_{mn})\|}$
- 8: Compute image-image attraction:
- 9: $A_I(i'_m) = \frac{E_I(i'_m) \cdot E_I(i_{ns})}{\|E_I(i'_m)\| \|E_I(i_{ns})\|}$
- 10: Compute semantic loss:
- 11: $\mathcal{L}_{sem} = [\lambda D_T(i'_m) - (1 - \lambda) A_I(i'_m)]_+$
- 12: Apply DWT to i_m and i'_m , and reconstruct low-frequency components $\bar{i}_m$ and $\bar{i}'_m$.
- 13: Compute frequency loss: $\mathcal{L}_{freq} = \|\bar{i}_m - \bar{i}'_m\|_1$
- 14: Update i'_m by minimizing: $\mathcal{L} = \mathcal{L}_{sem} + \beta \mathcal{L}_{freq}$
- 15: **until** convergence or maximum iterations K
- 16: Set $i_m^{adv} \leftarrow i'_m$.
- 17: **end for**

Output: Adversarial images $\{i_m^{adv}\}_{m=1}^M$.

b) Frequency-Domain Constraint for Imperceptible Perturbation: Pixel-domain adversarial perturbations often introduce visible artifacts, such as unnatural noise or structural distortion. To improve imperceptibility, we constrain the attack in the frequency domain by exploiting the different sensitivity of the human visual system (HVS) to frequency components. Since the HVS is more sensitive to low-frequency information that preserves global structure and less sensitive to high-frequency details such as textures and local noise [24], [60], [61], we encourage adversarial perturbations to concentrate mainly on high-frequency bands.

We employ two-dimensional Discrete Wavelet Transform (DWT) for spectral decomposition. As shown in Fig. 2, DWT decomposes an image i_m into one low-frequency approximation sub-band and three high-frequency detail sub-bands:

$$\begin{aligned} i_m^{ll} &= L i_m L^T, & i_m^{lh} &= H i_m L^T, \\ i_m^{hl} &= L i_m H^T, & i_m^{hh} &= H i_m H^T. \end{aligned} \quad (9)$$

Here, L and H denote low-pass and high-pass filters, respectively. The low-frequency component i_m^{ll} preserves global structure, while i_m^{lh} , i_m^{hl} , and i_m^{hh} capture horizontal, vertical, and diagonal details. Therefore, restricting changes in i_m^{ll} helps preserve perceptual quality.

Specifically, we reconstruct the low-frequency component of the original image as

$$\bar{i}_m = L^T i_m^{ll} L = L^T (L i_m L^T) L. \quad (10)$$

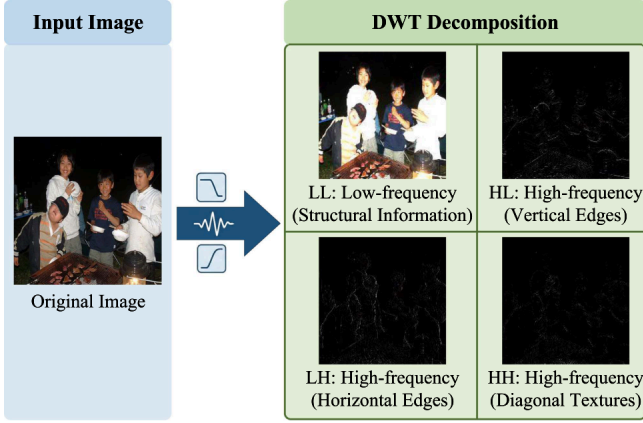


Fig. 2. Multiscale Image Decomposition via Discrete Wavelet Transform (DWT). The DWT decomposes a representative input image into four distinct components: the low-frequency (LL), which encapsulates the core structural information through lossy approximation, and three high-frequency (HL, LH, HH) that collectively characterize textural features. Specifically, (1) LL: Preserves compressed approximation of the original image through low-pass filtering in both spatial dimensions, retaining essential structural information while allowing controlled data reduction. (2) HL: Extracts vertical edge features via vertical high-pass and horizontal low-pass filtering operations. (3) LH: Identifies horizontal edge characteristics through horizontal high-pass and vertical low-pass filtering. (4) HH: Captures diagonal texture patterns by applying high-pass filters along both spatial axes.

For an intermediate adversarial image i'_m , its low-frequency reconstruction $\tilde{i}'_m$ is computed in the same way. The frequency-domain constraint is then defined as

$$\mathcal{L}_{freq}(i'_m) = \|\tilde{i}'_m - i'_m\|_1. \quad (11)$$

This loss penalizes low-frequency structural changes and encourages perturbations to remain in less perceptible high-frequency regions.

c) Integrated Optimization Framework: The final adversarial image i_m^{adv} is generated by jointly optimizing the dual-semantic attack objective and the frequency-domain constraint:

$$i_m^{adv} = \arg \min_{i'_m} (\mathcal{L}_{sem}(i'_m) + \beta \mathcal{L}_{freq}(i'_m)), \quad (12)$$

where

$$\mathcal{L}_{sem}(i'_m) = [\lambda D_T(i'_m) - (1 - \lambda) A_I(i'_m)]_+. \quad (13)$$

Here, $\mathcal{L}_{sem}$ weakens image-text alignment while guiding the image toward the non-similar visual target, and $\mathcal{L}_{freq}$ preserves low-frequency structure for visual stealth. The hyperparameter $\beta \geq 0$ balances attack strength and imperceptibility: a larger β better preserves structural fidelity, while a smaller β allows stronger semantic perturbation.

Overall, this objective enables DSFG-Attack to generate adversarial images that induce dual-semantic mismatch while maintaining high visual fidelity. The complete pipeline is illustrated in Fig. 1 and summarized in Algorithms 1 and 2.

V. EXPERIMENTS

In this section, we perform experimental analyses on VLP models using various datasets to demonstrate the advantages of DSFG-Attack. First, we present the experiment-related settings in Section V-A. Then, in Section V-B, we compare DSFG-Attack with similar methods and analyze the experimental results. Section V-C evaluates the cross-task attack performance of adversarial images. We further conduct experiments in Section V-D to examine whether the adversarial images generated based on the image text retrieval task can effectively target the currently popular MLLMs. Subsequently, extensive ablation studies are provided in Section V-E. Finally, we visualize multimodal adversarial examples, provide an in-depth stealthiness analysis of the generated adversarial texts, and discuss the feasibility and practicality of exposing the risks of DSFG-Attack.

A. Experimental Settings

1) Models: The evaluation framework encompasses both fused and aligned VLP architectures [13], [14], [36]. For the fused VLP model, we choose ALBEF and TCL with the same structure. Specifically, the unimodal image encoder consists of a vision transformer ViT-B/16, and the unimodal text encoder and multi-modal encoder both consist of a 6-layer transformer [25], [33], [35], [39]. For the aligned VLP model that contains only the unimodal image encoder and the unimodal text encoder [20], we consider CLIP for the evaluation and set ViT-B/16 and ResNet-101 as image encoders [34], [38], [39], [62], denoted as CLIP_{ViT} and CLIP_{CNN} , respectively.

2) Datasets: The benchmark evaluation of DSFG-Attack employs two multimodal datasets: Flickr30K [18] and MSCOCO [19]. Flickr30K contains 31,783 image-text pairs with five captions per image, while MSCOCO provides 123,287 pairs featuring five descriptive captions per visual instance. To ensure computational efficiency while maintaining statistical significance, we implement random stratified sampling to extract evaluation subsets (1,000 pairs from Flickr30K and 5,000 pairs from MSCOCO) [63]. This protocol enables systematic assessment of adversarial perturbations across diverse visual-semantic relationships while preserving dataset structure integrity.

3) Adversarial Attack Settings: For the comparison methods (i.e., Sep-Attack, Co-Attack [13], SGA [14], DRA [15]), we follow the parameter settings outlined in their respective papers. Specifically, for image modal perturbation, we set the perturbation boundary to $2/255$, the size of the step size to $0.5/255$, and the number of steps in the iteration to 10. We use the BERT-Attack [50] method for text modal adversarial attack with the perturbation boundary to 1 and the word list length set to 10. For DSFG-Attack, we set the batch size M to 8. When attacking image data, we use the Adam optimizer with a learning rate of 0.001, running for 100 iterations. Additionally, we set the hyperparameters λ and β to 0.9 and 1.2, respectively. For attacking text modality data, we also set the similarity threshold $\gamma = 0.7$, the perturbation boundary ε_t to 1 and the word list length to 10.

4) Evaluation Metric: We evaluate the robustness of the VLP model in image-text retrieval tasks, employing the Attack

TABLE I

COMPARE ATTACK SUCCESS RATES (%) WITH STATE-OF-THE-ART METHODS ON THE MSCOCO DATASET. WE SHOW EXPERIMENTAL RESULTS FOR R@1 ON IMAGE-TEXT RETRIEVAL. THE SOURCE MODEL (VERTICAL) GENERATES MULTI-MODAL ADVERSARIAL EXAMPLES TO ATTACK THE TARGET MODEL (HORIZONTAL). **NOTE:** * DENOTES WHITE-BOX ATTACK CONFIGURATIONS WITH FULL MODEL PARAMETER ACCESS.

Model	Attack	ALBEF		TCL		CLIP _{ViT}		CLIP _{CNN}	
		TR R@1	IR R@1	TR R@1	IR R@1	TR R@1	IR R@1	TR R@1	IR R@1
ALBEF	Sep-Attack	82.60*	89.88*	32.83	42.92	44.03	54.46	46.96	55.88
	Co-Attack	79.87*	87.83*	32.62	43.09	44.89	54.75	47.30	55.64
	SGA	96.75*	96.95*	58.56	65.38	57.06	65.25	58.95	66.52
	DRA	96.57*	96.47*	60.69	67.46	61.69	67.43	62.32	69.22
	DSFG-Attack	99.16*	98.90*	65.33	71.03	63.79	72.16	65.83	72.10
TCL	Sep-Attack	41.71	52.97	70.32*	78.97*	50.74	60.13	51.90	61.26
	Co-Attack	46.08	57.09	85.38	91.39*	51.62	60.46	52.13	62.49
	SGA	65.93	73.30	98.97*	99.15*	56.34	63.99	59.44	65.70
	DRA	68.06	75.86	98.99*	99.02*	63.30	68.59	64.24	70.58
	DSFG-Attack	71.55	76.76	99.11*	99.40*	68.18	73.41	71.41	76.03
CLIP _{ViT}	Sep-Attack	23.41	34.61	25.77	36.84	68.52*	77.94*	43.11	49.76
	Co-Attack	30.28	42.67	32.84	44.69	97.98*	98.80*	55.08	62.51
	SGA	33.41	44.64	37.54	47.76	99.79*	99.79*	58.93	65.83
	DRA	35.96	48.00	36.32	48.56	99.66*	99.70*	64.41	69.99
	DSFG-Attack	47.55	57.69	49.41	60.76	99.95*	99.99*	73.37	77.04
CLIP _{CNN}	Sep-Attack	26.53	39.29	30.26	41.51	50.44	57.11	88.72*	92.49*
	Co-Attack	29.83	41.97	32.97	43.72	53.10	58.90	96.72*	98.56*
	SGA	31.61	43.00	34.81	45.95	56.62	60.77	99.61*	99.80*
	DRA	33.26	45.15	33.89	46.49	59.60	64.87	99.51*	99.70*
	DSFG-Attack	45.71	57.02	48.10	59.91	69.30	71.72	99.93*	99.95*

Success Rate (ASR) as our evaluation metric. For instance, when using adversarial images as input to retrieve text data from a database, ASR refers to the percentage of retrieved erroneous text. Additionally, we **focus on** assessing adversarial images using four different metrics: average ℓ_2 distortion,¹ SSIM, CIEDE2000, and average distortion of low-frequency (LF) components. SSIM measures the structural similarity between adversarial and original images, CIEDE2000 assesses color differences, and LF analyzes the distortion level of the main components (low-frequency components) in adversarial images.

B. Experimental Results of DSFG-Attack

Note: As a pioneering work on frequency-domain perturbations for VLP models, DSFG-Attack lacks directly comparable baselines in existing literature. We therefore employ pixel-level perturbation methods for comparative analysis, while noting that DSFG-Attack operates with significantly lower perturbation intensity than these conventional approaches.

The transferability of adversarial examples is the primary metric for evaluating our attack in a practical black-box setting. In our experiments, we select a VLP model as the source model to generate multimodal adversarial examples and then evaluate their ASR on other, unseen target models. The comprehensive results presented in Tables I and II demonstrate the significant advantages of DSFG-Attack.

¹The ℓ_∞ norm has limited applicability for evaluating DSFG-Attack, as the perturbations generated by our method are primarily concentrated in image texture and edge regions, which are inherently stealth-prone.

Overall Attack Performance: First, we briefly summarize the white-box performance, where adversarial examples are generated and evaluated on the same model. As shown in Tables I, DSFG-Attack performs comparably to or even slightly surpasses strong baselines like SGA and DRA. For instance, when attacking the ALBEF model (indicated by *), DSFG-Attack improves the average ASR by 2.59% on Text Retrieval (TR) and 2.43% on Image Retrieval (IR) tasks, confirming its effectiveness when model gradients are accessible.

The core of our evaluation focuses on **black-box transferability**. We analyze the results from two critical dimensions: model type and model architecture.

a) Analysis by Model Type: DSFG-Attack demonstrates a pronounced advantage when transferring attacks between different types of VLP models (e.g., from a fusion-based model to an alignment-based model). For example, when using the fusion-based model ALBEF as the source, the transfer attack ASR against the alignment-based model CLIP_{ViT} surpasses the DRA baseline by an average of 3.42%. Similarly, using TCL as the source improves the attack on CLIP_{CNN} by an average of 6.31% over DRA. We attribute this superior cross-type transferability to our dual-semantic guidance mechanism. By explicitly pulling the adversarial image towards a dissimilar visual semantic anchor, we create a more generalized and model-agnostic semantic shift that persists across different model paradigms.

Notably, DSFG-Attack also provides substantial gains even when the source and target models are of the same type. When using the fusion-based model TCL to attack another fusion-based model ALBEF, DSFG-Attack achieves an average improvement of 2.2% over DRA. This indicates that the semantic guidance enhances perturbation coherence and transferability beyond simple cross-type scenarios.

TABLE II

COMPARE ATTACK SUCCESS RATES (%) WITH STATE-OF-THE-ART METHODS ON THE FLICKR30 K DATASET. WE SHOW EXPERIMENTAL RESULTS FOR R@1 ON IMAGE-TEXT RETRIEVAL. THE SOURCE MODEL (VERTICAL) GENERATES MULTI-MODAL ADVERSARIAL EXAMPLES TO ATTACK THE TARGET MODEL (HORIZONTAL). **NOTE:** * DENOTES WHITE-BOX ATTACK CONFIGURATIONS WITH FULL MODEL PARAMETER ACCESS.

Model	Attack	ALBEF		TCL		CLIP _{ViT}		CLIP _{CNN}	
		TR R@1	IR R@1	TR R@1	IR R@1	TR R@1	IR R@1	TR R@1	IR R@1
ALBEF	Sep-Attack	65.69*	73.95*	17.60	32.95	31.17	45.23	32.82	45.49
	Co-Attack	77.16*	83.86*	15.21	29.49	23.60	36.48	25.12	38.89
	SGA	97.24*	97.28*	45.42	55.25	33.38	44.16	34.93	46.57
	DRA	97.50*	96.63*	49.74	58.83	39.14	48.39	41.38	51.66
	DSFG-Attack	97.39*	97.66*	48.05	57.33	40.06	50.93	44.44	53.96
TCL	Sep-Attack	20.13	36.48	84.72*	86.07*	31.29	44.65	33.33	45.80
	Co-Attack	23.15	40.04	77.94*	85.59*	27.85	41.19	30.74	44.11
	SGA	48.91	60.34	98.37*	98.81*	33.87	44.88	37.74	48.30
	DRA	51.09	61.79	98.21*	98.33*	40.25	48.94	42.91	52.49
	DSFG-Attack	49.99	60.02	97.65*	97.96*	43.93	55.42	45.88	57.98
CLIP _{ViT}	Sep-Attack	9.59	23.25	11.38	25.60	79.75*	86.79*	30.78	39.76
	Co-Attack	10.57	24.33	11.94	26.69	93.25*	95.86*	32.52	41.82
	SGA	13.40	27.22	16.23	30.76	99.08*	98.94*	38.76	47.79
	DRA	12.51	30.00	14.65	30.62	98.77*	99.00*	45.47	50.74
	DSFG-Attack	18.13	38.11	22.04	40.21	99.90*	99.95*	50.88	58.33
CLIP _{CNN}	Sep-Attack	8.55	23.41	12.64	26.12	28.34	39.43	91.44*	95.44*
	Co-Attack	8.79	23.74	13.10	26.07	28.79	40.03	94.76*	96.89*
	SGA	11.42	24.80	14.91	28.82	31.24	42.12	99.24*	99.49*
	DRA	12.20	26.59	14.33	29.29	35.21	45.94	99.11*	99.49*
	DSFG-Attack	19.47	37.88	21.78	40.48	40.40	52.20	100.00*	99.97*

b) Analysis by Model Architecture: We further dissect transferability by examining the image encoder architecture within the alignment-based CLIP model, testing both Vision Transformer (ViT) and Convolutional Neural Network (CNN) backbones. A clear trend emerges from Table II: adversarial examples consistently transfer more successfully from ViT-based encoders to CNN-based encoders than vice versa. For instance, when CLIP_{ViT} is the source, the attack on CLIP_{CNN} achieves 50.88% (TR R@1) and 58.33% (IR R@1). In the reverse direction (CLIP_{CNN} → CLIP_{ViT}), the ASR drops to 40.40% and 52.20%, respectively. This asymmetric transferability, also observed when ALBEF is the source (where attacks on CLIP_{CNN} are 3.71% higher on average than on CLIP_{ViT}), suggests that ViT-based encoders possess inherent robustness advantages over CNNs against the studied adversarial perturbations, likely due to the global receptive field and self-attention mechanisms in Transformers.

c) Quantitative Stealth Analysis of Adversarial Images: Beyond ASR, we employ four quantitative metrics to rigorously assess the imperceptibility of our adversarial images, comparing against the SGA baseline [14] under varying pixel perturbation budgets (2/255, 8/255, 16/255). The results are summarized in Table IV.

- *Perturbation Magnitude (ℓ_2 -norm):* Our method produces perturbations with significantly lower ℓ_2 -norm than the baselines. With $\lambda = 0.9$, our ℓ_2 -norm is 4.32 units lower than a strong baseline, indicating superior stealth.
- *Structural Similarity (SSIM):* DSFG-Attack consistently achieves an SSIM score of 0.99 across different λ values, matching the high similarity scores of the baselines and confirming that structural information is well-preserved.

- *Color Difference (CIEDE2000):* The CIEDE2000 metric shows a strong negative correlation with λ . At $\lambda = 0.9$, our score is 125.98 units lower than the baseline, demonstrating that our frequency-domain constraint effectively minimizes perceptible color distortion.
- *Low-Frequency Distortion (LF):* The LF metric (which quantifies changes to fundamental image structure) is markedly lower for DSFG-Attack than for all baseline budgets. This provides direct evidence that our perturbations are successfully confined to high-frequency components, aligning with the design goal of our frequency constraint.

In summary, DSFG-Attack not only establishes new state-of-the-art black-box transferability across diverse VLP models and architectures but also does so while generating adversarial images that are quantitatively superior in terms of visual stealth compared to pixel-domain attack methods.

C. Cross-Task Transferability

To further assess the generalization capability of the adversarial examples generated by DSFG-Attack, we evaluate their transferability across different vision-language tasks. Specifically, we investigate whether adversarial images crafted to disrupt image-text retrieval can also degrade performance on the distinct but related task of image captionings [30], [64], [65]. Image captioning requires a model to generate coherent and semantically accurate textual descriptions from visual input, a process typically involving an image encoder followed by a text decoder [66]. Successful cross-task transfer would indicate that our attack targets fundamental, task-agnostic visual-semantic representations within VLP models.

TABLE III
CROSS-TASK ATTACKS. WE LEVERAGE MSCOCO TO GENERATE
ADVERSARIAL IMAGES FROM ALBEF ON THE IMAGE-TEXT RETRIEVAL TASK,
ATTACKING THE BLIP MODEL FOR THE IMAGE CAPTIONING TASK.

Attack	B@4 ↓	METEOR ↓	ROUGE_L ↓	CIDEr ↓	SPICE ↓
Baseline	39.7	31.0	60.0	133.3	23.8
Co-Attack	37.4	29.8	58.4	125.5	22.8
SGA	34.8	28.4	56.3	116.0	21.4
DRA	35.3	28.4	56.5	117.2	21.6
DSFG-Attack	34.6	28.1	56.0	114.8	20.9

In our experiment, we use the ALBEF model, pre-trained on MSCOCO for image-text retrieval, as the source model to generate adversarial images via DSFG-Attack. These adversarial images are then used to attack the BLIP model [25], which is specifically designed for the target task of image captioning. The quality of the generated captions is measured using five standard metrics: BLEU, METEOR, ROUGE_L, CIDEr, and SPICE.

As shown in Table III, when the frequency constraint parameter β is set to 0.2, DSFG-Attack causes a consistent degradation across all five evaluation metrics compared to the baseline. This demonstrates its significant cross-task attack capability. A detailed analysis reveals that, relative to the baseline, DSFG-Attack reduces the BLEU-4 (B@4) score by 0.7%, indicating notable lexical deviations between the adversarial and reference captions. More importantly, the CIDEr score, which measures semantic consensus, shows a more pronounced decline of 2.4%. This highlights that the adversarial perturbations effectively reduce the correlation between the generated captions and the ground-truth references, thereby undermining the semantic fidelity of the captioning model's output.

These results confirm that the adversarial vulnerabilities exploited by DSFG-Attack are not confined to a single task objective but generalize to other downstream applications that rely on the same foundational visual understanding. The perturbations induce a generalized semantic distortion that persists across different model architectures and task-specific heads.

D. Adversarial Transferability in MLLMs

Given the growing prominence and deployment of Multimodal Large Language Models (MLLMs), assessing their robustness against adversarial attacks has become a critical research concern. To this end, we extend our evaluation to investigate whether the adversarial examples generated by DSFG-Attack can transfer to and compromise state-of-the-art MLLMs. This experiment probes the existence of shared vulnerabilities in visual feature representations across different model families and scales.

In recent years, MLLMs have garnered significant attention, sparking widespread interest among researchers. Consequently, we began investigating their robustness in the context of adversarial attacks. Specifically, we used ALBEF as the source model to generate adversarial images (setting parameter β to 0) targeting MLLMs: the latest and stable GPT-4o (standard version, closed-source model)² [26] and Qwen 2 (open-source

model)³ [27]. A standardized prompt (“Please describe this image, limited to 50 words.”) elicited model responses, with Figs. 4 and 5 documenting critical failure modes through red-highlighted outputs. For example, in the first pair of Fig. 4, we queried GPT-4o with an adversarial image, which incorrectly identified a standing person, while the actual scene clearly depicted three people clustered together. In the second pair of samples, GPT-4o mistakenly recognized “crocodile” as “dinosaur”. In Fig. 5, for the first sample pair, Qwen 2 incorrectly identifies a person holding a microphone. In the second sample pair, Qwen 2 fails to successfully recognize that two individuals are engaged in a fighting situation. These transfer-based adversarial successes demonstrate that parameter-agnostic attacks can compromise even state-of-the-art MLLMs through cross-model vulnerability propagation.

E. Ablation Study

We conduct a series of ablation experiments to systematically evaluate the individual contributions of key components in DSFG-Attack and to determine optimal hyperparameter settings. The analysis focuses on three critical aspects: 1) the balancing coefficient λ governing the dual-semantic guidance, 2) the frequency constraint weight β controlling perturbation imperceptibility, 3) the influence of different inter-modal interaction strategies on attack efficacy, 4) comparative analysis of the performance impact of image perturbations under pixel-level and frequency-domain constraints, 5) comparative analysis of image-text matching across models under single-pair and dual-semantic guidance approaches, and 6) the attack performance and stealthiness of non-similar target selection compared with random target selection.

1) *Analysis of the Dual-Semantic Balancing Coefficient λ* : The hyperparameter $\lambda \in (0, 1]$ balances the two components of our dual-semantic guidance loss: the text-image divergence D_T and the image-image attraction A_I . It effectively controls the relative influence of repulsion from the adversarial text and attraction towards the non-similar visual target.

We evaluate the impact of λ by sweeping its value from 0.1 to 1.0 in increments of 0.1, measuring the ASR in both white-box and black-box transfer scenarios. As illustrated in Fig. 3, the white-box ASR shows a pronounced increasing trend within the interval [0.1, 0.8] and plateaus thereafter. This indicates that a stronger emphasis on repelling the adversarial text (higher λ) is highly effective when the model's gradients are accessible. Conversely, for black-box transferability, the ASR exhibits a more nuanced behavior: it increases slightly as λ rises from 0.1 to 0.9, but begins to decline as λ approaches 1.0. This suggests that over-emphasizing text repulsion at the complete expense of visual attraction ($\lambda = 1$) can degrade the semantic coherence of the perturbation, thereby harming its transferability to unseen models.

To complement the ASR analysis, we quantify the perceptual stealth of adversarial images generated with different λ values. As shown in Table IV, we report standard image quality

²GPT-4o: <https://chatgpt.com/>

³Qwen 2: <https://huggingface.co/spaces?sort=trending&search=qwen>

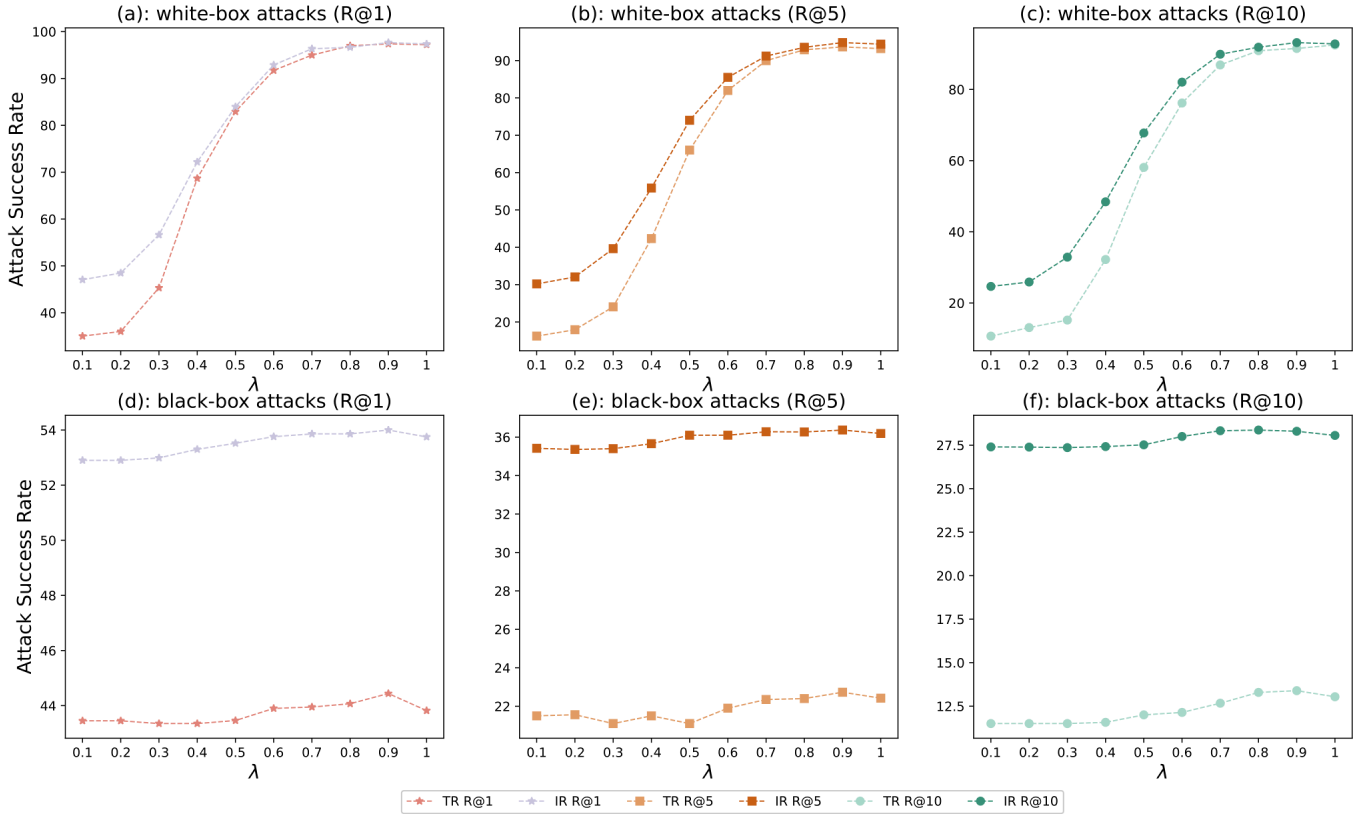


Fig. 3. Effect of parameter λ on the attack success rate within image-text retrieval task. We generate adversarial examples for Flickr30K on ALBEF, attacking the target model CLIP_{CNN}. The first column shows white-box attacks from R@1 to R@10, and the second column shows black-box attacks from R@1 to R@10.

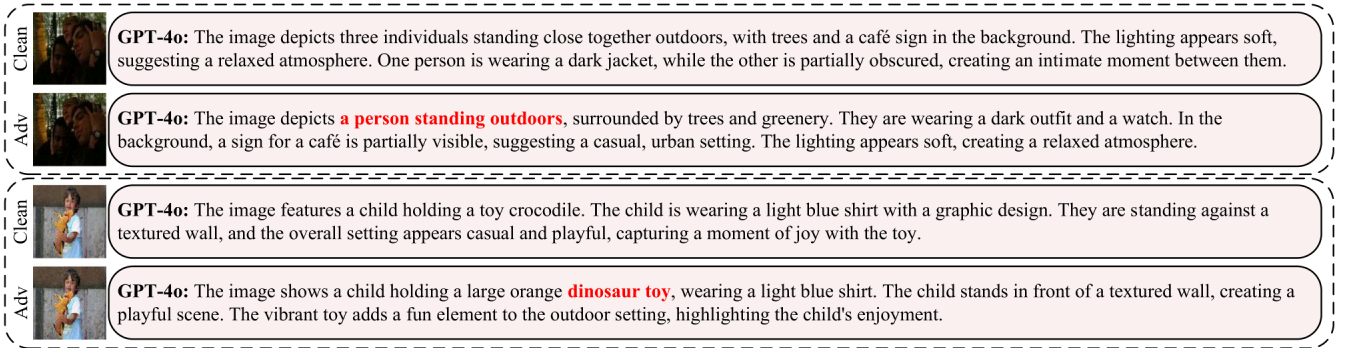


Fig. 4. Cross-model adversarial attack demonstration using DSFG-Attack on GPT-4o (standard version). The dashed boxes compare original images (*i.e.*, Clean) with baseline responses and ALBEF-generated adversarial images (*i.e.*, Adv) with corrupted outputs. Critical failures are annotated in red: (1) human count error (3 persons misidentified as 1) and (2) toy category confusion (crocodile misclassified as dinosaur). All responses were generated under the standardized instruction: Please describe this image, limited to 50 words.

metrics (including the ℓ_2 -norm of the perturbation, the Structural Similarity Index (SSIM), the CIEDE2000 color difference, and a Low-Frequency (LF) similarity measure) under three perturbation budgets for a baseline method (SGA [14]). The results indicate that lower λ values consistently yield smaller ℓ_2 , CIEDE2000, and LF deviations, while SSIM remains close to 1. This demonstrates that reducing λ (*i.e.*, increasing the weight of visual attraction) enhances the imperceptibility of the adversarial perturbation. For instance, with $\lambda = 0.1$, the ℓ_2 -norm

is a negligible 0.01, compared to 779.19 for a strong baseline perturbation (budget 16/255).

Balancing the white-box performance and black-box transferability, we select $\lambda = 0.9$ as the default setting. This value achieves near-optimal white-box ASR while maintaining robust black-box transfer performance.

2) *Analysis of the Frequency Constraint Weight β* : The hyperparameter $\beta \geq 0$ controls the strength of the frequency-domain constraint $\mathcal{L}_{freq}$, which is crucial for enforcing the

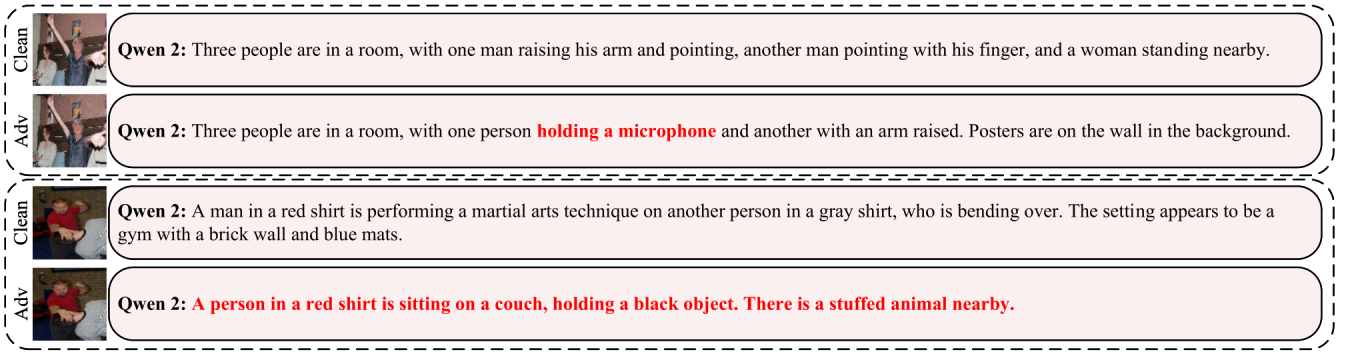


Fig. 5. Cross-model adversarial attack demonstration using DSFG-Attack on Qwen 2. The dashed boxes compare original images (*i.e.*, Clean) with baseline responses and ALBEF-generated adversarial images (*i.e.*, Adv) with corrupted outputs. Critical failures are annotated in red: (1) handheld object misclassification (tool misidentified as microphone) and (2) scene recognition error (combat scenario mislabeled as casual interaction). All responses were generated under the standardized instruction: Please describe this image, limited to 50 words.

TABLE IV
FOUR METRICS (*I.E.*, ℓ_2 , SSIM, CIEDE2000 AND LF) RELATED TO THE STEALTHINESS OF ADVERSARIAL IMAGES UNDER DIFFERENT PARAMETERS λ . TO ESTABLISH COMPREHENSIVE PERFORMANCE BASELINES, THREE MAXIMUM PERTURBATION BUDGETS (2/255, 8/255, 16/255) WERE IMPLEMENTED FOR THE PIXEL-LEVEL SGA [14]. WE EMPLOYED THE ALBEF TO GENERATE ADVERSARIAL IMAGE ON THE FLICKR30K DATASET, WITH ALL METRIC VALUES DERIVED THROUGH COMPUTATIONAL ANALYSIS AGAINST CLEAR IMAGES.

λ	$\ell_2 \downarrow$	SSIM $\uparrow$	CIEDE2000 $\downarrow$	LF $\downarrow$
Baseline 1 (16/255)	779.19	0.7	3074.81	393.37
Baseline 2 (8/255)	223.85	0.87	1727.77	124.06
Baseline 3 (2/255)	17.34	0.99	544.33	10.2178
1.0	14.16	0.99	437.03	0.0048
0.9	13.02	0.99	418.35	0.0035
0.8	11.34	0.99	400.23	0.0036
0.7	9.32	0.99	379.77	0.0037
0.6	7.22	0.99	345.28	0.0036
0.5	5.12	0.99	298.72	0.0038
0.4	2.58	0.99	244.63	0.0033
0.3	1.34	0.99	197.34	0.0029
0.2	0.93	0.99	160.34	0.0007
0.1	0.02	1.00	142.77	0.0001

visual imperceptibility of image perturbations. A larger β penalizes modifications to the low-frequency structural information more heavily, confining perturbations to less perceptible high-frequency bands.

We investigate the effect of β by testing a range of values: $\{0, 0.2, 0.4, 0.6, 0.8, 1.0, 1.2, 1.5, 2.0, 3.0, 5.0\}$. The black-box ASR on the CLIP_{CNN} model are presented in Table V. The quantitative analysis reveals a clear trade-off: as β increases, the ASR gradually decreases, while the perceptual quality of the adversarial images improves. Specifically, for $\beta > 1.2$, the white-box attack performance begins to degrade, whereas the black-box transferability remains relatively stable. In terms of stealth metrics (ℓ_2 , SSIM, CIEDE2000, LF), significant improvement in imperceptibility is observed once β exceeds 0.2. The limited perceptual divergence shown in Fig. 2 is a direct result of this frequency-constrained optimization, which strategically concentrates modifications on high-frequency components

(*e.g.*, fine textures and edges) to which the human visual system is less sensitive.

Based on this trade-off analysis, we select $\beta = 1.2$ as the default value. This setting provides a favorable balance, maintaining strong attack potency while ensuring high visual fidelity of the adversarial images.

3) *Analysis of Inter-Modal Interaction Strategies*: The sequential interaction between image and text modalities is a core design element of our pipeline. We investigate how varying the number and order of interaction stages affects the transferability of the generated adversarial examples. Specifically, we evaluate six interaction strategies of increasing complexity: text-image (t-i), image-text (i-t), image-text-image (i-t-i), text-image-text (t-i-t), text-image-text-image (t-i-t-i), and text-image-text-image-text (t-i-t-i-t).

In these multi-stage strategies, each "image" or "text" step denotes one complete adversarial generation cycle for that modality, using the output of the previous step as guidance. For example, the four-stage t-i-t-i strategy operates as follows: (1) original images guide the generation of initial adversarial text; (2) this adversarial text guides the generation of adversarial images; (3) these updated images guide a second refinement of the adversarial text; (4) the refined text finally guides the generation of the ultimate adversarial images.

The ASR for transferring adversarial examples from a CLIP_{ViT} source model to a CLIP_{CNN} target model under these strategies are summarized in Fig. 6. The results indicate that the simple two-stage t-i interaction achieves strong transferability. While more complex, iterative strategies (involving three or more stages) can yield marginal improvements in some cases, these gains are often outweighed by the significant increase in computational cost. Moreover, excessive iteration may lead to over-adaptation to the source model, potentially harming transferability. Therefore, DSFG-Attack adopts the minimal effective two-stage text-image (t-i) interaction strategy, which optimally balances attack efficacy with operational efficiency.

4) *Pixel-Level Vs. Frequency-Domain Constraints*: To conduct an in-depth analysis of the DSFG-Attack method, we perform a comparative study between two constraint strategies: Pixel-level and Frequency-domain. Specifically, we replace the

TABLE V

THE IMPACT OF PARAMETER β ON FIVE METRICS (I.E., ASR, ℓ_2 , SSIM, CIEDE2000 AND LF) WITHIN IMAGE-TEXT RETRIEVAL TASK. QUANTITATIVE MEASUREMENTS ACROSS β CONFIGURATIONS ARE PRESENTED IN TABLE, WITH COMPLEMENTARY QUALITATIVE ANALYSIS ENABLED THROUGH FIGURE 7'S VISUAL COMPARISON OF ADVERSARIAL PERTURBATIONS. WE GENERATE ADVERSARIAL EXAMPLES FOR FLICKR30K ON ALBEF, ATTACKING THE TARGET MODEL CLIP_{CNN}. NOTE: * DENOTES WHITE-BOX ATTACK CONFIGURATIONS WITH FULL MODEL PARAMETER ACCESS.

β	ALBEF*(%)						CLIP _{CNN} (%)						$\ell_2 \downarrow$	SSIM $\uparrow$	CIEDE2000 $\downarrow$	LF $\downarrow$
	TR			IR			TR			IR						
	R@1	R@5	R@10	R@1	R@5	R@10	R@1	R@5	R@10	R@1	R@5	R@10				
0	98.64	97.09	96.10	98.90	97.52	96.52	46.10	24.42	14.21	55.51	37.77	29.77	23.92	0.99	608.97	12.3993
0.2	97.50	93.89	91.90	97.66	94.81	93.17	43.55	22.94	12.98	53.76	36.49	28.26	16.91	0.99	434.08	0.0072
0.4	97.39	93.69	91.50	97.57	94.63	92.88	43.55	22.94	12.98	53.96	36.17	28.35	15.97	0.99	433.09	0.0037
0.6	96.98	93.89	91.50	97.38	94.71	92.94	43.81	22.94	12.87	53.89	36.39	28.12	15.06	0.99	429.93	0.0033
0.8	97.08	93.69	91.50	97.33	94.61	92.96	44.06	22.94	13.18	53.96	36.34	28.35	14.29	0.99	426.21	0.0034
1.0	97.08	93.29	91.60	97.31	94.61	92.76	44.19	22.62	13.29	54.00	36.27	28.46	13.61	0.99	422.19	0.0035
1.2	97.39	93.42	91.53	97.66	94.59	92.78	44.44	22.73	13.39	53.96	36.37	28.26	13.02	0.99	418.35	0.0035
1.5	97.08	93.19	91.30	97.17	94.28	92.68	43.93	22.52	13.49	54.03	36.27	28.37	12.27	0.99	412.93	0.0037
2.0	96.77	92.79	91.00	97.03	94.13	92.50	43.81	22.52	13.70	54.10	36.44	28.26	11.29	0.99	404.82	0.0040
3.0	96.45	92.28	90.20	96.70	93.77	92.07	43.55	22.20	13.49	53.96	36.24	28.37	9.90	0.99	391.60	0.0046
5.0	96.04	90.98	89.30	96.19	93.23	91.37	43.55	22.73	12.98	53.72	36.32	28.26	8.15	0.99	371.64	0.0056

TABLE VI

IMPACT OF IMAGE CONSTRAINTS (PIXEL-LEVEL VS. FREQUENCY-DOMAIN) ON FIVE EVALUATION METRICS (ASR, ℓ_2 , SSIM, CIEDE2000, AND LF) IN IMAGE-TEXT RETRIEVAL. ADVERSARIAL EXAMPLES ARE GENERATED ON FLICKR30K USING CLIP_{CNN} AND TRANSFERRED TO TCL (* DENOTES WHITE-BOX ATTACK WITH FULL MODEL ACCESS).

Disruption Type	CLIP _{CNN} *(%)						TCL(%)						$\ell_2 \downarrow$	SSIM $\uparrow$	CIEDE2000 $\downarrow$	LF $\downarrow$
	TR			IR			TR			IR						
	R@1	R@5	R@10	R@1	R@5	R@10	R@1	R@5	R@10	R@1	R@5	R@10				
Pixel-level	99.01*	97.79*	96.15*	99.28*	98.57*	97.51*	22.51	7.41	4.35	40.51	24.52	18.71	16.87	0.99	543.82	10.2096
Frequency-domain	100.00*	99.90*	98.98*	99.97*	99.76*	99.48*	21.78	7.79	4.16	40.48	24.63	18.73	12.68	0.99	412.76	0.0033

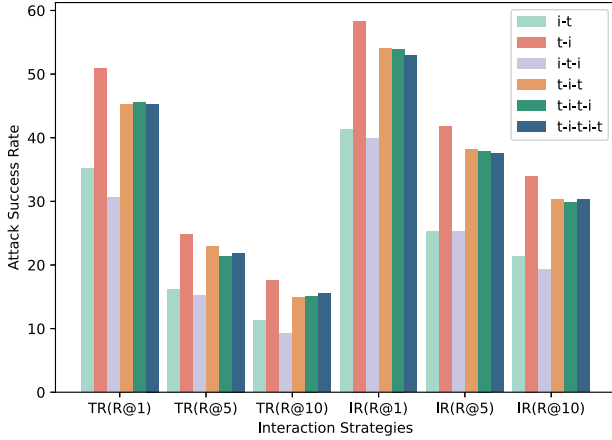


Fig. 6. Attack success rate (%) with different interaction strategies. We generate adversarial examples for Flickr30K on CLIP_{ViT}, attacking the target model CLIP_{CNN}.

frequency-domain constraint in DSFG-Attack with a conventional pixel-level perturbation while keeping all other settings unchanged. The maximum perturbation magnitude for images is set to $2/255$. The experimental results are presented in Table VI, which compares five evaluation metrics under the two constraints: ASR, ℓ_2 distortion, SSIM, CIEDE2000, and average distortion of LF components.

In terms of attack efficacy, the frequency-domain constraint exhibits a marginal decrease in black-box ASR compared to the pixel-level approach, a trade-off attributable to its more restrictive perturbation budget designed for enhanced stealth.

Conversely, under white-box conditions, the frequency-domain constraint demonstrates superior and more stable performance across all recall levels (R@1 to R@10). Next, regarding perceptual quality, the frequency-domain constraint leads to significant improvements in ℓ_2 distortion, CIEDE2000, and LF distortion, while maintaining a comparable level of SSIM to the pixel-level approach.

In summary, by adopting the frequency-domain constraint, DSFG-Attack achieves substantially enhanced stealthiness with only a marginal reduction in black-box attack performance, alongside a notable improvement in white-box effectiveness. This configuration offers a more balanced trade-off and helps fill a gap in stealth-aware adversarial attack design for VLP models.

5) *Single-Pair Vs. Dual-Semantic Guidance*: To highlight the advantages of the dual-semantic guidance in DSFG-Attack, we perform a comparative analysis between two semantic guidance strategies: Single-Pair and Dual-Semantic Guidance. The evaluation focuses on image-text matching scores, as cross-modal alignment is central to the image-text retrieval task studied in this work. Specifically, we replace the Single-Pair data used in the baseline method with our Dual-Semantic Guidance while keeping all other experimental settings unchanged, and analyze the resulting changes in matching scores.

As shown in Table VII, we present three randomly selected image-text pairs to illustrate how matching scores vary when shifting from Single-Pair to Dual-Semantic Guidance. For example, in the first pair, the adversarial example generated by SGA under the Single-Pair setting yields a matching score of 0.1623, which decreases to 0.1502 under Dual-Semantic

TABLE VII

ALIGNMENT SCORES OF ADVERSARIAL IMAGES ACROSS MODELS. SCORES ARE LOWEST FOR THE SOURCE MODEL ALBEF (*), SHOW SLIGHT INCREASES IN THE TARGET MODELS (TCL, CLIP_{ViT} AND CLIP_{CNN}), AND DECLINE UNDER DUAL-SEMANTIC GUIDANCE.

No.	Clear Text	Model	Degree of Alignment with Adversarial Image (Single-Pair → Dual-Semantic Guidance)
1	“tons of baskets of apples for sale at an outside market.”	ALBEF*	0.1623* → 0.1502*
		TCL	0.1965 → 0.1833
		CLIP _{ViT}	0.3152 → 0.2968
		CLIP _{CNN}	0.2857 → 0.2617
2	“a small pink beanie hat next to a cell phone.”	ALBEF*	0.1891* → 0.1651*
		TCL	0.2235 → 0.2051
		CLIP _{ViT}	0.3164 → 0.2857
		CLIP _{CNN}	0.2966 → 0.2712
3	“a close up of a number of zebras on a dirt ground.”	ALBEF*	0.2129* → 0.1946*
		TCL	0.2462 → 0.2311
		CLIP _{ViT}	0.3043 → 0.2772
		CLIP _{CNN}	0.2859 → 0.2604

TABLE VIII

EFFECT OF TARGET SELECTION STRATEGY ON BLACK-BOX TRANSFERABILITY (ALBEF → CLIP_{CNN}, FLICKR30K, * DENOTES WHITE-BOX ATTACK WITH FULL MODEL ACCESS)

Target Selection	ALBEF(%)		CLIP _{CNN} (%)		$\ell_2 \downarrow$	SSIM $\uparrow$	CIEDE2000 $\downarrow$	LF $\downarrow$
	TR R@1	IR R@1	TR R@1	IR R@1				
Random	96.96±0.37*	97.17±0.38*	44.03±0.18	53.90±0.14	13.74±0.14	0.99±0	430.78±11.998	0.0043±0.0007
Non-similar	97.39*	97.66*	44.44	53.96	13.02	0.99	418.35	0.0035

Guidance (a reduction of 0.0121). These results consistently demonstrate that the proposed dual-semantic guidance further disrupts cross-modal alignment, thereby validating its effectiveness in degrading retrieval performance.

6) *Non-Similar Vs. Random Target Selection*: To further validate that the dissimilarity of the selected target image i_{ns} is critical for transferability, we compare our non-similar target selection strategy against baselines: **random target**, where we randomly pick an image i_{rand} from the same mini-batch ($j \neq m$) without considering semantic similarity.

We conduct this ablation on the Flickr30K dataset, using ALBEF as the source model and attacking the black-box target CLIP_{CNN}. All other hyperparameters ($\lambda = 0.9$, $\beta = 1.2$, batch size $M = 8$) remain identical to our default setting. The random target experiment is repeated 5 times, and we report the average ASR for TR and IR at R@1.

As shown in Table VIII, the non-similar target strategy achieves higher and more stable black-box ASR, while maintaining better stealthiness. Specifically, it obtains 44.44% for TR and 53.96% for IR, compared with 44.03% and 53.90% achieved by random target selection. Although the variance of the random baseline is small under the current controlled setting, its performance depends on the sampled target and may fluctuate more in open scenarios or larger mini-batches [67]. These results indicate that actively selecting a semantically dissimilar anchor provides more stable guidance and promotes a more model-agnostic semantic drift, thereby improving black-box transferability.

F. Qualitative Analysis and Visualization

We present a qualitative evaluation of the adversarial examples generated by DSFG-Attack through a visual comparison of six randomly selected clean and adversarial image-text pairs,

as shown in Fig. 7. This analysis serves to verify two key design properties: the imperceptibility of the perturbations and the semantic-preserving nature of the textual attacks.

a) *Visual Imperceptibility*: The adversarial pairs are visually indistinguishable from their original counterparts under standard viewing conditions. This stealth is achieved through targeted, minimal perturbations in each modality. For the image modality, the third row of Fig. 7 visualizes the imperceptible noise by displaying the pixel-wise residual magnified 50 times. A critical observation is that these amplified perturbations are predominantly concentrated in high-frequency regions, while the primary low-frequency structural contours of objects remain intact. This pattern provides empirical validation for our frequency-domain constraint, confirming its success in confining adversarial modifications to less perceptible spectral components.

b) *Semantic-Preserving Textual Attack*: For the text modality, adversariality is introduced via precise, single-word substitutions, highlighted in red. This demonstrates the efficiency of our token-localization strategy in identifying impactful yet minimal lexical changes.

The stealth of our textual attack extends beyond surface-level appearance to semantic coherence. This is enforced by our candidate replacement mechanism, which selects substitutes from a constrained semantic neighborhood within the GloVe embedding space, requiring high cosine similarity to the original word. Consequently, perturbations preserve lexical meaning and contextual integrity (e.g., “little” → “small”).

c) *Embedding-Level Attack vs. Word-Level Attack*: To evaluate the stealth and semantic coherence of adversarial text generation, we compare our proposed Embedding-Level Attack with a representative Word-Level Attack (BERT-Attack [50]). Table IX presents four representative adversarial examples generated by each method.

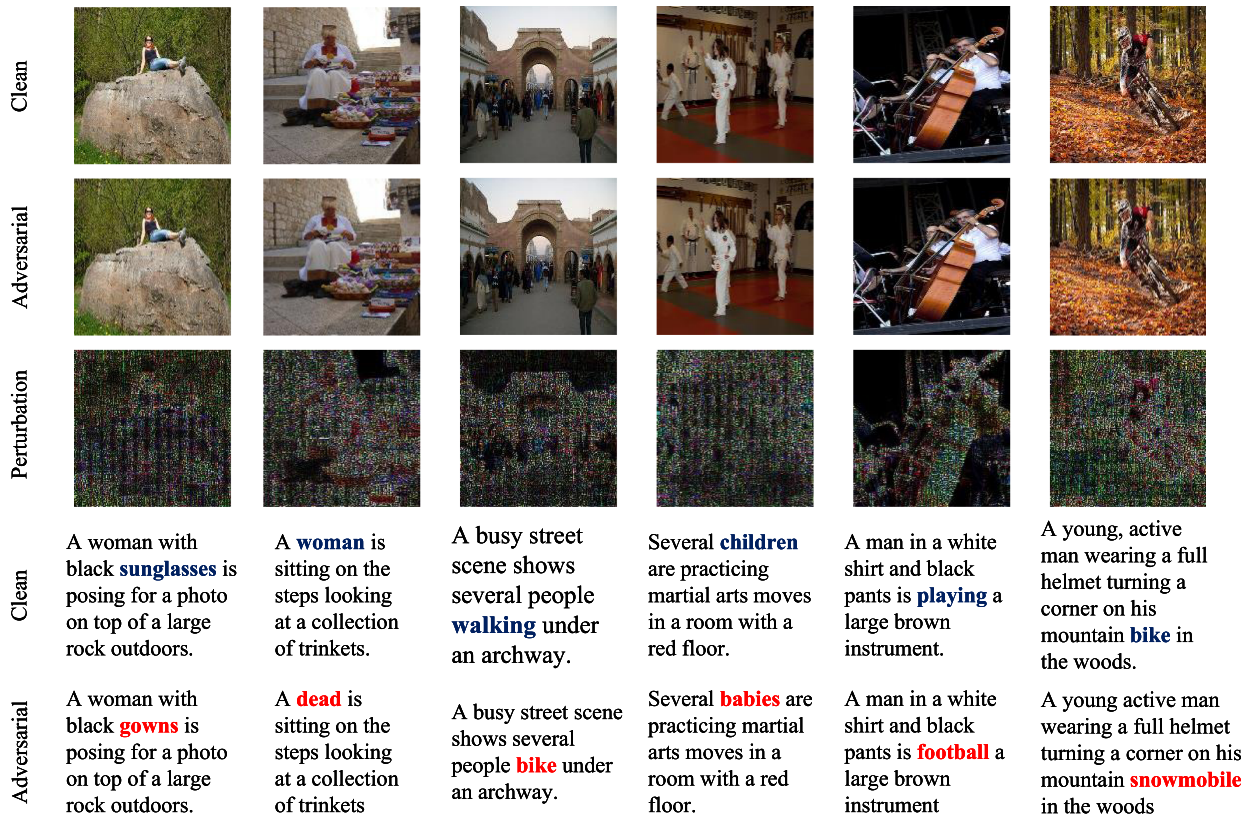


Fig. 7. Visualization of DSFG-Attack. We randomly selected six image-text pairs from the Flickr30K dataset and used ALBEF to generate adversarial examples. The first row presents the original images, the second row displays the adversarial images generated by DSFG-Attack, and the third row shows the perturbations magnified 50 times for enhanced visualization. The fourth row contains the unaltered text, while the final row shows the text with applied perturbations.

TABLE IX

COMPARISON OF ADVERSARIAL TEXTS GENERATED BY THE EMBEDDING-LEVEL ATTACK AND THE WORD-LEVEL ATTACK (BERT-ATTACK). THE SECOND COLUMN PRESENTS THE ORIGINAL CLEAN CAPTIONS. THE THIRD COLUMN SHOWS ADVERSARIAL TEXTS PRODUCED BY THE WORD-LEVEL ATTACK. THE FOURTH COLUMN DISPLAYS ADVERSARIAL TEXTS GENERATED BY THE PROPOSED EMBEDDING-LEVEL ATTACK. PERTURBED TOKENS ARE HIGHLIGHTED IN RED.

No.	Clear Text	Word-Level Attack (BERT-Attack)	Embedding-Level Attack (GloVe)
1	"A young girl rollerblades, while protected by many pads and pillows."	"A young girl girl . . while protected by many pads and pillows."	"A young girl carpooling while protected by many pads and pillows"
2	"Asian people in a temple praying over a smokey urn."	"Asian people in a temple praying over a smokey - flame. "	" Africa people in a temple praying over a smokey urn"
3	"Three young kids enjoying them self at the community park."	"... young kids enjoying them self at the community park."	" Fifteen young kids enjoying them self at the community park."
4	"A lady with dark hair is playing a harp."	"A lady with dark ... is playing a harp."	"A lady with dark hair is playing a shakuhachi. "

The outputs of the Word-Level Attack exhibit notable artifacts that undermine imperceptibility. These include lexical repetitions (No.1) and the unnatural insertion of underscores and ellipses in irregular positions (No.1–4), such as sentences beginning with an ellipsis. Such patterns deviate significantly from standard grammatical and stylistic conventions, rendering the adversarial text easily detectable.

In contrast, our Embedding-Level Attack enforces a semantic similarity constraint within a continuous embedding space. By restricting candidate replacements to tokens exceeding a pre-defined cosine similarity threshold (0.7 in our experiments), the method ensures that substitutions remain within the semantic neighborhood of the original word. This approach largely avoids generating linguistically anomalous constructs (such as

misplaced punctuation or semantically incongruent terms) by leveraging the structured representation of the GloVe vocabulary. For example, as shown in Table IX, our method replaces "Asian" with "Africa" and "Three" with "Fifteen". These substitutions preserve contextual relevance while effectively altering the semantic content, demonstrating the ability of the embedding-space search to maintain textual plausibility during adversarial perturbation.

G. Feasibility and Practicality of the Revealed Risk

To better justify the practical feasibility of the revealed risk, we clarify the operational assumptions of DSFG-Attack. In the black-box setting, the attacker does not require access to the

target model's architecture, parameters, gradients, or training data. Instead, adversarial examples are generated on a surrogate VLP model and then directly transferred to unseen target models. This assumption is realistic because many VLP models, such as CLIP-like aligned models and ALBEF/TCL-like fusion models, have publicly available implementations or can be approximated by open-source surrogates. The required input is also practical: DSFG-Attack only uses ordinary image-text pairs and a small mini-batch to select a non-similar semantic anchor. It does not require additional annotation or target-model queries during perturbation generation.

From a computational perspective, the attack is feasible because the textual perturbation is performed through constrained candidate search in the embedding space, while the image perturbation is optimized with a fixed number of iterations on the surrogate model. More importantly, the attack is designed to be stealthy: image perturbations are constrained mainly in high-frequency components, and textual substitutions are restricted to semantically related candidates. Therefore, the generated adversarial examples are difficult to detect visually or linguistically. The feasibility of this threat is empirically supported by our black-box results in Tables I–II, where adversarial examples generated from one source model consistently degrade multiple unseen target models with different architectures. Furthermore, the cross-task results in Table III and the MLLM case studies in Figs. 4–5 show that the generated perturbations can transfer beyond the original retrieval task and affect advanced multimodal systems. These results indicate that DSFG-Attack exposes a practical and transferable risk rather than an attack limited to an idealized white-box setting.

VI. LIMITATIONS

While DSFG-Attack demonstrates strong adversarial performance across multiple VLP models, its generalizability is constrained in specific respects. Most notably, the approach shows limited cross-task transferability; adversarial examples optimized for image-text retrieval exhibit reduced effectiveness when applied to other tasks, such as visual grounding. We identify two contributing factors. First, the framework is prone to **task-specific overfitting**, as optimization guided by retrieval-specific losses yields perturbations that do not generalize well to other multimodal objectives. Second, a **spectral sensitivity disparity** arises: models designed for different tasks may rely on features that are less susceptible to the frequency-domain perturbations tailored for retrieval models. These limitations point to the need for future research into more generalizable adversarial formulations that can produce task-agnostic perturbations capable of compromising diverse multimodal systems.

VII. CONCLUSION

This paper presents DSFG-Attack, a multimodal adversarial attack framework that enhances transferability through dual-semantic guidance and improves stealth via frequency-domain and semantic constraints. Unlike approaches that merely amplify semantic differences on the source model, our method introduces dissimilar semantics to maintain adversarial divergence

under black-box transfer. Adversarial stealth is strengthened by restricting image perturbations to high-frequency components and constraining text perturbations within the GloVe embedding space.

Evaluations on Flickr30K and MSCOCO show that DSFG-Attack achieves superior transferability across multiple VLP models, confirming the value of dual-semantic alignment. By jointly optimizing for low perceptibility and high transferability, our method obtains state-of-the-art results in stealthy multimodal adversarial attacks.

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Haiqi Zhang received the MS degree from Guangxi Normal University, Guilin, China, in 2022. He is currently working toward the PhD degree with the School of Computer Science and Engineering, Nanjing University of Science and Technology, Nanjing, China. His research interests include multi-modal adversarial attack and multi-modal learning.



Ziqiang Li received the BE degree from the University of Science and Technology of China (USTC), Hefei, China, in 2019, and the PhD degree from the University of Science and Technology of China (USTC), Hefei, China, in 2024. He is currently a lecturer with the Engineering Research Center of Digital Forensics, Ministry of Education, Nanjing University of Information Science and Technology. His research interests include deep generative models, AI Security, and machine learning.



Hao Tang (Member, IEEE) received the PhD degree in computer science and technology from the Nanjing University of Science and Technology, Nanjing, China, in 2024. He is currently a postdoctoral fellow with the Centre for Smart Health, The Hong Kong Polytechnic University. He has authored or coauthored more than 30 technical papers in top journals and conferences, including three that have been recognized as ESI highly cited or hot papers. His research interests include machine learning, computer vision, and medical image analysis, focusing on fine-grained

learning, data-efficient learning, and multi-modal learning, and their practical applications.



Zechao Li (Senior Member, IEEE) received the BE degree from the University of Science and Technology of China, in 2008, and the PhD degree from the National Laboratory of Pattern Recognition, Institute of Automation, Chinese Academy of Sciences, in 2013. He is currently a professor with the Nanjing University of Science and Technology. His research interests include big media analysis and computer vision. He was the recipient of the Best Paper Award at ACM Multimedia Asia 2020 and 2024. He is also an associate editor for *IEEE Transactions on Pattern*

Analysis and Machine Intelligence, *IEEE Transactions on Neural Networks and Learning Systems*, *IEEE Transactions on Multimedia*, and *IEEE Transactions on Circuits and Systems for Video Technology*.